SECTION 1 IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY

1.1 Product identifiers

Product Form	Substance
Trade Name	Lithium Iron Phosphate (LiFePO4) Powder for Li-ion Battery Cathode Application
Product Number	NG08BE1404
CAS Number	15365-14-7

1.2 Relevant identified uses of the substance or mixture and uses adviced against

Use of the substance/mixture	Research

1.3 Details of the supplier of safety data sheet

Company	Nanografi Nanotechnology
Address	ODTÜ Teknokent İkizler Binası B-1/H ODTÜ Teknokent 06531- ANKARA
Phone	+90 312 285 85 09
Fax	+90 312 210 13 09

1.4 Emergency Telephone Number

Emergency Telephone Number	+90 312 285 85 09

SECTION 2 HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

Not a hazardous substance or mixture according to Regulation (EC) No. 1272/2008.

2.2 Label Elements

Hazard Pictograms	Not available
Signal Word	Not available

Hazard Statements	Not available
Precautionary Statements	Not available

Not a hazardous substance or mixture according to Regulation (EC) No. 1272/2008.

2.3 Other Hazards

None.

SECTION 3 COMPOSITION/INFORMATION ON INGREDIENTS

3.1 Substances

Synonyms	Lithium Iron Phosphate, Iron Lithium Phosphate, Ferrous Lithium Phosphate, LFP
Formula	LiFePO4
CAS Number	15365-14-7

3.2 Mixtures

Not applicable.

SECTION 4 FIRST AID MEASURES

4.1 Description of first aid measures

First-aid measures after inhalation	Move person into fresh air. If not breathing give artificial respiration.
First-aid measures after skin contact	Wash off with soap and plenty of water.
First-aid measures after eye contact	Flush eyes with water as a precaution.
First-aid measures after ingestion	Never give anything by mouth to an unconscious person. Rinse mouth with water.

4.2 Most important symptoms and effects, including acute and delayed

No additional information available.

4.3 Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

4.4 Other Information

No data available.

SECTION 5 FIREFIGHTING MEASURES

5.1 Extinguishing media

Suitable Extinguishing Media	Water mist, water spray, Carbon Dioxide, dry chemical foam.

5.2 Special hazards arising from the substance or mixture

Phosphorus oxides. Lithium oxides. Iron oxides.

5.3 Advice for firefighters

Wear self-contained breathing apparatus if necessary.

SECTION 6 ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

6.1.1 For non-emergency personnel

Emergency Procedures	Avoid dust formation. Avoid breathing vapors, mist or gas.

6.2.2 For emergency responders

Protective Equipment	Do not attempt to take action without suitable protective equipment. For further information refer to section 8: "Exposure controls/personal protection".

6.2 Environmental precautions

No data available.

6.3 Methods and material for containment and cleaning up

Avoid raising dust. Use mechanical cleaning equipment. Keep in suitable, closed containers for disposal.

6.4 Reference to other sections

For further information refer to section 13.

SECTION 7 HANDLING AND STORAGE

7.1 Precautions for safe handling

Precautions for Safe Handling	Ensure good ventilation of the work station.
Hygiene measures	Always wash hands after handling the product.

7.2 Conditions for safe storage, including any incompatibilities

Storage Conditions	Store in dry, well ventilated areas.

7.3 Specific end use(s)

Apart from the uses mentioned in section 1.2 no other specific uses are stipulated.

SECTION 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Contains no substances with occupational exposure limits.

8.2 Exposure controls

Appropriate Engineering Controls	General industrial hygiene practice.
Hand Protection	Wear regular work gloves.

Eye Protection	Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).
Skin and Body Protection	The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.
Respiratory Protection	Respiratory protection is not required. Where protection from nuisance levels of dusts are desired, use type N95 (US) or type P1 (EN 143) dust masks. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).
Environmental Exposure Controls	No special environmental precautions required.

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Physical State	Solid
Appearance	Powder
Odor	Odorless
Odor Threshold	No data available
pH	No data available
Relative Evaporation Rate	No data available
Melting Point	>300 °C
Freezing Point	No data available
Boiling Point	No data available
Flash Point	No data available
Auto-ignition Temperature	No data available
Decomposition Temperature	No data available
Flammability (solid, gas)	No data available
Vapor Pressure	No data available
Relative vapor density at 20°C	No data available
Relative Density	No data available

Solubility in H2O	No data available
Log Pow	No data available
Viscosity, Kinematic	No data available
Viscosity, Dynamic	No data available
Explosive Properties	No data available
Oxidizing Properties	No data available

9.2 Other information

No other information available.

SECTION 10 STABILITY AND REACTIVITY

10.1 Reactivity

No data available.

10.2 Chemical stability

Stable at normal temperatures.

10.3 Possibility of hazardous reactions

No data available.

10.4 Conditions to avoid

No data available.

10.5 Incompatible materials

Strong oxidizing agents.

10.6 Hazardous decomposition products

Thermal decomposition may release Phosphorus oxides, lithium oxides, iron oxides.

SECTION 11 TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Acute Toxicity	Not classified
Skin corrosion/irritation	Not classified
Serious eye damage/irritation	Not classified
Respiratory or Skin Sensitization	Not classified
Germ Cell Mutagenicity	Not classified
Carcinogenity	No component of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed human carcinogen by IARC. No component of this product present at levels greater than or equal to 0.1% is identified as a known or anticipated carcinogen by NTP. No component of this product present at levels greater than or equal to 0.1% is identified as a carcinogen or potential carcinogen by OSHA.
Reproductive Toxicity	Not classified
STOT-Single Exposure	Not classified
STOT-Repeated Exposure	Not classified
Aspiration Hazard	Not classified
Other Information	To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

SECTION 12 ECOLOGICAL INFORMATION

12.1 Toxicity

Ecology-general	No data available.
Acute aquatic toxicity	No data available.
Chronic aquatic toxicity	No data available.

12.2 Persistance and degradability

No data available.

12.3 Bioaccumulative potential

No data available.

12.4 Mobility in soil

No data available.

12.5 Results of PBT and vPvB assessment

PBT/vPvB assessment not available as chemical safety assessment not required/not conducted.

12.6 Other adverse effects

No data available.

SECTION 13 DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Product	Offer surplus and non-recyclable solutions to a licensed disposal company.
Contaminated Packaging	Dispose of as unused product.

SECTION 14 TRANSPORT INFORMATION

Safe to carry by any means of transportation. Not restricted any mode of transportation.

SECTION 15 REGULATORY INFORMATION

15.1 Safety, health and environmental regulations

The product does not need to be labelled in accordance with Directive 1999/45/EC, or Annex VI to 67/548/EEC. Not a hazardous substance or preparation according to EC-directives 67/548/EEC or 1999/45/EC.

SECTION 16 OTHER INFORMATION

16.1 Abbreviation of acronyms

ADR	European Agreement concerning the International Carriage of Dangerous Goods by Road
CLP	Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008
IATA	International Air Transport Association
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals Regulation (EC) No 1907/2006
PBT	Persistent Bioaccumulative Toxic
RID	Regulations concerning the International Carriage of Dangerous Goods by Rail
vPvB	Very Persistent and Very Bioaccumulative

16.2 Full text of H-statements

Not applicable.

16.3 Hazard classification

Health	0
Flammability	0
Reactivity	0
Special Hazards	None as per National Fire Protection Association (NFPA). The material should not be used in medical applications involving permanent implantation in the human body.

Disclaimer

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore beconstrued as guaranteeing any specific property of the product.